

Dillon Blyth

Dundee, UK | 07341 251430 | dillonrblyth29@gmail.com | **GitHub:** github.com/Akuto2904 | **Portfolio:** akuto2904.github.io/Portfolio-Website/

Personal Profile

BSc Computer Science student at the University of Dundee currently averaging First-Class/2:1 Honours equivalent. Highly interested in full-stack software engineering, AI model efficiency, and system optimization. Seeking a Year in Industry/Summer Internship to apply rigorous problem-solving skills to scalable, real-world products.

Key Skills

- **Languages:** C++, Python, Java, C, JavaScript, SQL, HTML/CSS, Bash.
- **Frameworks & Tools:** PyTorch, React, Node.js, Git, GitHub Pages, Linux, CI/CD, MongoDB, MySQL.
- **Core Domains:** Data Structures & Algorithms, AI Quantization, Graph Theory, DB Design.

Education

BSc (Hons) Computer Science | University of Dundee | *Sept 2024 – Jun 2028 (Expected)*

- **Grade:** Averaging A3–B2 (First-Class / 2:1 Honours Equivalent).
- **Coursework:** Data Structures & Algorithms, Systems Architecture, Software Engineering, Web Dev.

High School Qualifications | Woodmill High School | *Aug 2022 – Jun 2024*

- **Advanced Higher:** Chemistry | **Highers:** Comp Sci, Maths, Physics, Human Biology, English | **National 5's:** Applications of math

Technical Projects

AI Warehouse Grid Pathfinding Agent | *Python, PyTorch*

- Engineered a deep learning AI model using PyTorch to navigate grid environments and compute optimal paths for automated warehouse logistics.
- Implemented spatial constraint navigation and grid-based optimization to enhance decision logic.

Interactive Team Web Application | *Flutter, React, Agile*

- Built a multi-framework interactive website within a cross-functional team, utilizing Git version control and dynamic state management for streamlined CI/CD pipelines.

Sports Coaching Database Architectures | *MongoDB, MySQL*

- Architected a multi-collection NoSQL MongoDB platform with role-based access and aggregation pipelines for dynamic analytics (e.g., revenue, ratings).
- Designed a relational MySQL database for multi-facility management, executing dynamic table relationships and constraint setup scripts.

Artist Portfolio Website | *JavaScript, HTML/CSS, Bootstrap*

- Developed an intuitive, responsive website for an independent artist, featuring a dynamic client contact form and deployed seamlessly via GitHub Pages.

Unix To-Do List CLI | *Bash, Linux (Ubuntu)*

- Built a Unix command-line task manager and calendar system using Bash scripting to automate workflows, formatted with standard man page styling.

Experience & Leadership

Chef | Pacamara Food and Drink & ESS Compass | *Jul 2025 – Present*

- Co-manage fast-paced kitchen operations, executing 30+ meals/hour during peak service and acting as the lead point of contact in the head chef's absence.
- Manage £3,000 weekly inventory turnover, balancing 20 hours/week of employment alongside full-time STEM studies.

Badminton Club Convenor & Team Player | University of Dundee | *Sept 2024 – Present*

- Organize university-wide tournaments and dynamic ladder competitions for 200+ members, managing court logistics and sports union liaisons for 50 active competitors.

Teaching Assistant & Peer Tutor | Woodmill High School | *Aug 2022 – Jun 2024*

- Mentored ~60 students weekly across STEM subjects, breaking down complex algorithmic and mathematical concepts into accessible steps.

References available upon request.